EX N0:09 **FORWARD CHAINING** **Roll.no:241001051**

Exercise 1: Simple Inference

```
:-dynamic known/1.
%Initial knowledge
start :- assert(known(a)), assert(known(b)), infer.
%Rule
infer :- known(a), known(b), \+ known(c), assert(known(c)), write(&amp;#39;Inferred
c&amp;#39;), nl.
% Query
% ?- start.
```

Output:
Inferredc

Exercise 2: Fire Alert System

```
:-dynamic known/1.
%Rule to infer fire
check :- known(smoke), known(heat), assert(known(fire)), write(&amp;#39;Fire
Alert!&amp;#39;), nl.
% Start rule
start :- assert(known(smoke)), assert(known(heat)), check.
% Query
% ?- start
```

Output:
FireAlert!

Exercise 3: Water Purity Detection

```
:-dynamic known/1.
%Rule to infer pollution
infer_pollution :- known(cloudy_water), known(bad_smell), assert(known(polluted)),
write(&amp;#39;Water is polluted&amp;#39;), nl.
start :- assert(known(cloudy_water)), assert(known(bad_smell)), infer_pollution.
% Query
% ?- start
```

Output:
Waterispolluted

Result:

Hence,theb process of performing the forward chaining operations is verified successfully.